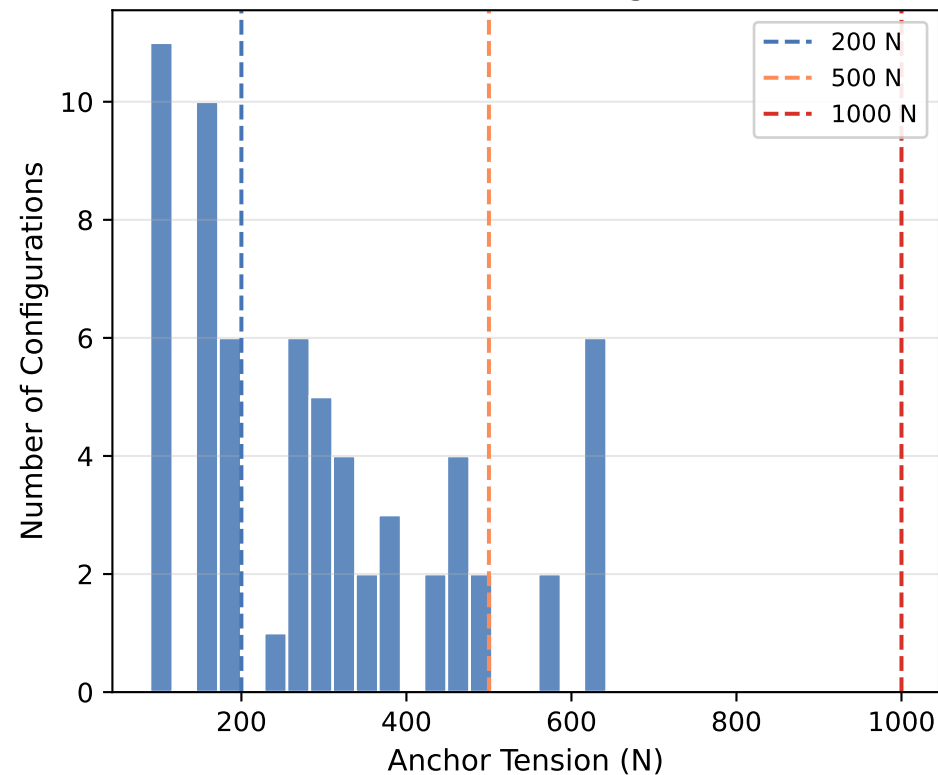
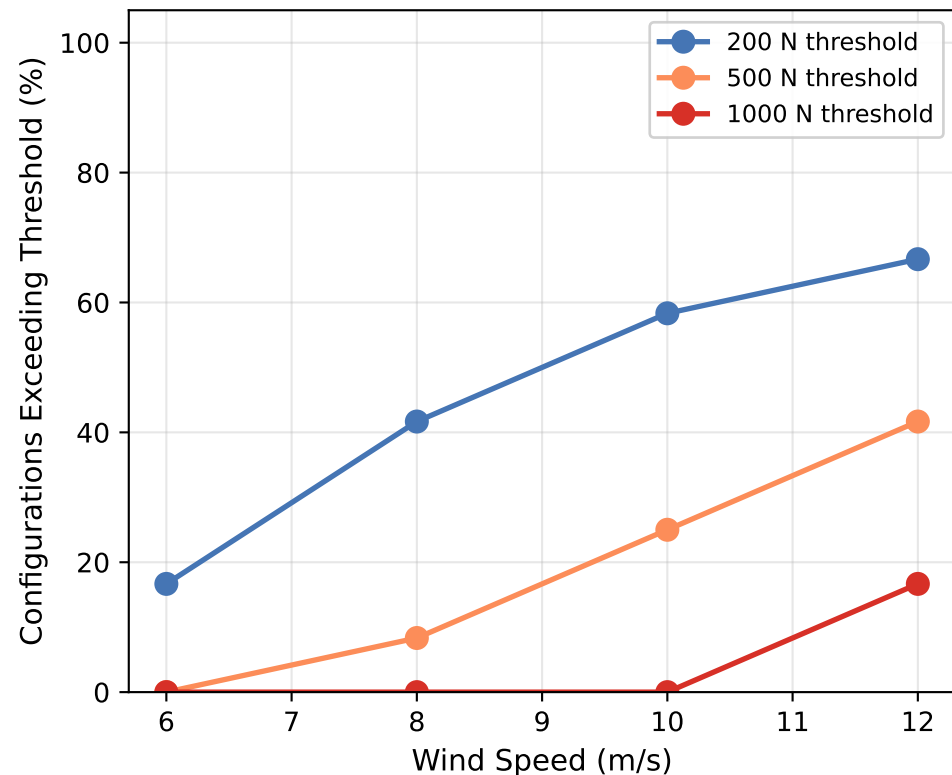


Capacity Factor: Useful Tension Across the Wind Spectrum

BEM v2.1 — All Profiles, 45° + 55° Elevation (all viable configs)

Tension Distribution at 8 m/s



IMPLICATIONS: At 6 m/s only 17% of configurations exceed 200 N. At 12 m/s, 92% exceed 500 N. The median tension at 8 m/s is 250 N. A Weibull $k=2$ distribution at 8 m/s mean gives ~65% capacity factor for the 200 N threshold. Larger rotors ($R=3.0m$) dominate the right tail. Data: BEM v2.1, 96 unique ($R,N,profile,elevation$) combinations.